

Charles Dalton

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EDUCATION

University of Florida

B.S. Nuclear Engineering & B.S. Physics

Aug 2023 – May 2027

GPA: 3.90/4.00

RESEARCH EXPERIENCE

Oak Ridge National Laboratory

Research Intern (DOE SULI) — Radioisotope Science and Technology Division

May 2026 – Present

Oak Ridge, TN

- Characterized double-pulse laser-induced breakdown spectroscopy (DP-LIBS) of flowing argon across a 720-configuration full factorial design (interpulse delay, pulse energies, flow rate, gate delay) as a candidate technique for real-time noble-gas monitoring in molten salt reactor off-gas streams.
- Operated a collinear DP-LIBS system (two 1,064 nm Nd:YAG lasers, six-channel spectrometer, ICCD plasma imaging) with mass flow control for four Ar I/Ar II analytical lines.
- Demonstrated $\sim 3\times$ neutral-line and $>10\times$ ionized-line SNR enhancement for DP over single-pulse operation, with neutral and ionized lines optimizing under opposing timing conditions.
- Built a Python analysis pipeline linking plasma morphology from ICCD imaging to spectral signal quality, using factorial ANOVA with cluster-robust inference to identify physical predictors of signal-to-noise ratio.
- Demonstrated that double-pulse operation buffers signal quality against flow-rate variability relative to single-pulse, supporting deployment in dynamic off-gas environments.

Hartig Lab, University of Florida

Undergraduate Research Assistant

Dec 2025 – Present

Gainesville, FL

- Conduct optical diagnostics (LIBS, emission imaging) of materials in nuclear-relevant environments, including excited-species population analysis and plume vorticity characterization.
- Develop spectral analysis methods including Voigt-profile line fitting with robustness safeguards for low-SNR conditions.

North Carolina State University

Research Assistant (DOE-Funded REU)

May 2025 – Jul 2025

Raleigh, NC

- Designed and built a custom Sample Analysis by Light Transmission (SALT) furnace for UV-Vis and THz-TDS spectroscopy of molten salts.
- Configured the optical system and conducted UV-Vis spectroscopy on molten LiCl-KCl eutectic and nitrate salts in support of a NEUP-funded modeling effort.

Jung Lab, University of Florida

Undergraduate Research Assistant

Jan 2024 – Aug 2025

Gainesville, FL

- Performed GC-MS experimentation on manufactured plastics and environmental samples to characterize polymer degradation and microplastic composition.

PUBLICATIONS (IN PREPARATION)

Dalton, C., Kitzhaber, Z., Hartig, K., & Andrews, H. “Double-pulse laser-induced breakdown spectroscopy of flowing argon for molten salt reactor off-gas monitoring.” Manuscript in preparation for *Spectrochimica Acta Part B: Atomic Spectroscopy*.

Borrero, J., Compitiello, A., Realmuto, E., Cook, J., Strack, A., **Dalton, C.**, & Hartig, K. “Spatiotemporal Imaging of Ground-State Al and AlO Reveals Oxygen-Source-Dependent Laser-Ablation Plume Chemistry.” Manuscript in preparation for *ACS Omega*.

PRESENTATIONS

Dalton, C., Kitzhaber, Z. B., & Andrews, H. B. “Configuration Optimization of Double-Pulse Laser-Induced Breakdown Spectroscopy in Flowing Argon.” *Oak Ridge National Laboratory SULI Summer Poster Session*, Oak Ridge, TN, July 2026. Poster.

Dalton, C. & Bataller, A. “Furnace Design and STrAW Method.” *Advancing Fundamental Molten Salt Modeling Using Ultrafast Spectroscopy, NEUP Annual Review Meeting*, Raleigh, NC, July 2025. Oral presentation.

Sedio, S., Paulson, D., Ritchie, J., **Dalton, C.**, & Jung, S. “Investigation Into the Effects of Location on Microplastic Composition on the University of Florida Campus.” *UF Undergraduate Research Symposium*, April 2024. Poster.

Dalton, C. & Magomo, D. “Smooth Transition Functions in Multiple Dimensions.” *Florida Undergraduate Research Conference*, Miami Gardens, FL, February 2023. Poster.

Dalton, C. & Magomo, D. “Smooth Transition Functions in Multiple Dimensions.” *Mathematical Association of America Conference*, Lakeland, FL, December 2022. Oral presentation.

LEADERSHIP & OUTREACH

Institute of Nuclear Materials Management, UF Student Chapter — Officer **Aug 2026 – Present**

- Lead the chapter officer team in planning technical events and outreach in nuclear materials management and safeguards.

American Nuclear Society, UF Chapter — Mentor **Aug 2025 – Present**

- Provide sustained academic and research guidance to nuclear engineering mentees; placed mentees into faculty research groups over the academic year.

Society of Physics Students — Outreach Volunteer **Aug 2025 – Present**

- Promote physics education through hands-on demonstrations at Alachua County public elementary schools.

HONORS & AWARDS

NEUP Research & Development Award (\$10,000) **2025**

- U.S. Department of Energy Nuclear Energy University Program award supporting university-led nuclear energy research.

University Scholars Program (\$1,750) **2025**

- Competitive University of Florida award funding a year of mentored undergraduate research.

UF Honors University Research Scholars Program Scholarship (\$2,000) **2025**

- Honors program scholarship recognizing sustained engagement in undergraduate research.

Florida Bright Futures Academic Scholars (FAS) Scholarship (~\$21,887) **2023 – Present**

- Florida’s highest-tier state merit scholarship, covering full tuition and applicable fees (120 credit hours).

SKILLS & RELEVANT COURSEWORK

Experimental: single- and double-pulse LIBS, ICCD imaging, echelle spectroscopy, UV–Vis spectroscopy, GC–MS, optical alignment, laser safety, mass flow control, CAD

Analysis & Programming: Python (NumPy, SciPy, pandas), MATLAB, C++, LaTeX, Git; spectral line fitting (Voigt profiles), factorial experiment design, ANOVA with cluster-robust inference, bootstrap methods

Coursework: Quantum Mechanics, Electrodynamics, Statistical Mechanics, Solid State Physics, Thermodynamics, Fluid Dynamics, Heat Transfer, Statics, Mechanics of Materials